



Produced-Water Lithium Extraction

TBAC/DA DES + TOPO from Smackover feed basis to staged extraction and screening TEA

Chemical engineering technical review

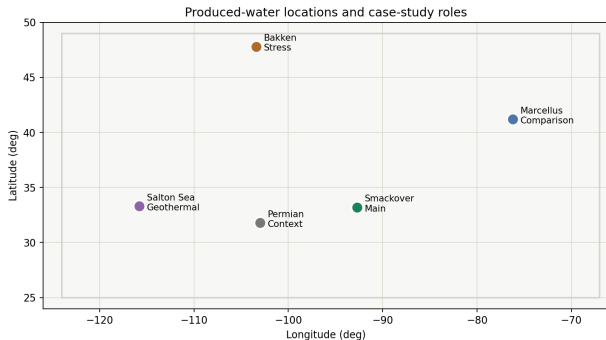
Smackover Li/Na feed basis → pretreatment → TBAC/DA DES + TOPO → ALAMO surrogate → PrOMMiS/IDAES

May 15, 2026

A pretreated Smackover Li/Na stream connects realistic produced-water chemistry to staged extraction and screening-level economics.

CASE BASIS

Smackover Is The Main Produced-Water Case



Case basis

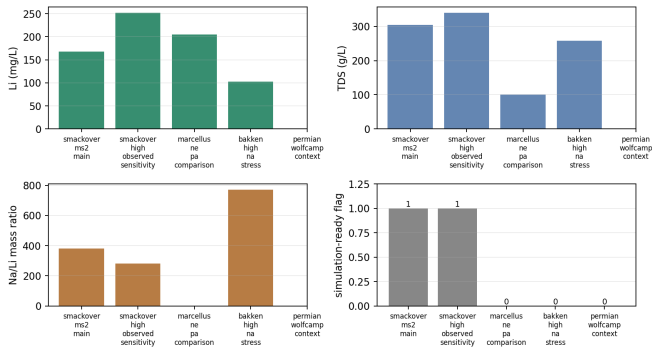
Southern Arkansas Smackover combines meaningful lithium grade with high-salinity major-ion chemistry.

- MS-2 feed: 168 mg/L Li, 64 100 mg/L Na, and 305 000 mg/L TDS.
- High-observed Smackover sensitivity: 252 mg/L Li.
- Marcellus, Bakken, Permian, and Salton Sea remain comparison cases.

Sources: USGS Smackover data release; selected feed table.

Feed Variance Sets The Model Role

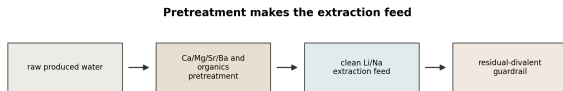
Feed variance drives model role and process credibility



- Smackover MS-2 anchors the main flowsheet and TEA case.
- The high-Li Smackover row tests feed-grade sensitivity.
- Incomplete major-ion rows stay outside the main process calculation.

Feed	Role
MS-2	main case
High-Li Smackover	sensitivity
Marcellus/Bakken	context

Pretreatment Creates The Li/Na Extraction Feed



- Raw produced water is not the extraction feed.
- Ca^{2+} , Mg^{2+} , Sr^{2+} , Ba^{2+} , suspended solids, oil, and organics are upstream treatment burdens.
- Lithium loss in pretreatment is carried into TEA sensitivity.

SOLVENT AND TRANSFER SURFACE

TBAC/DA DES + TOPO Is The Active Solvent

Fixed solvent formulation and Li/Na transfer role



- Organic phase: 90 wt% TBAC/DA DES and 10 wt% TOPO.
- TBAC:DA molar ratio: 1:2.
- Temperature: 23 °C.
- Aqueous pH: 10.4.

Source: Rezaee DES/TOPO extraction basis and supporting information.

Stage-Response Surface Spans Source And Smackover Domains

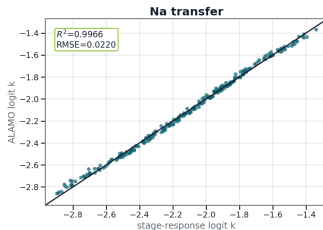
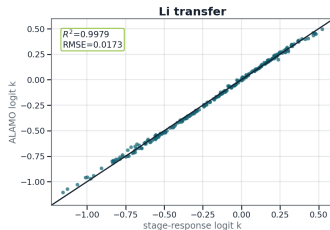
Domain	Rows	Use
source LHS	625	extraction anchors
Smackover LHS	625	feed sensitivity
reported anchors	3	Li/Na targets
nominal/corner rows	18	stress bounds

Retained source anchors

optimized Li extraction	48.57%
optimized Li/Na selectivity	4.41
model-brine Li extraction	51.63%
model-brine Na extraction	9.97%

- Reported extraction responses define the transfer anchors.
- Feed chemistry selects the Smackover case and validity bounds.
- The process calculation reports this paper-anchor transfer basis.

IDAES ALAMO Surrogate Preserves Li/Na Transfer Trends



- Inputs: Li concentration, Na concentration, O/A ratio, and TOPO loading.
- Outputs: logit-transformed Li and Na transfer coefficients.
- $R^2 = 0.9979$ for Li and $R^2 = 0.9966$ for Na on the held-out table.

Output	RMSE	max error
logit k_{Li}	0.0173	0.0555
logit k_{Na}	0.0220	0.0779

Paper-Response Variables Feed The Process Model

$$D_i = \frac{E_i}{100 - E_i} \frac{1}{O/A}$$

$$S_{\text{Li/Na}} = \frac{D_{\text{Li}}}{D_{\text{Na}}}$$

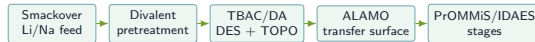
Rezaee anchors, $O/A = 1$

Optimized 10 wt% TOPO: $E_{\text{Li}} = 48.57\%$,

$S_{\text{Li/Na}} = 4.41$, so $E_{\text{Na}} = 11.01\%$.

Model-brine row: $E_{\text{Li}} = 51.63\%$ and

$E_{\text{Na}} = 9.97\%$.



Rezaee extraction responses are converted to Li/Na transfer variables for process calculations

PROCESS AND ECONOMICS

PrOMMiS/IDAES Uses Active Li And Na Transfer Terms

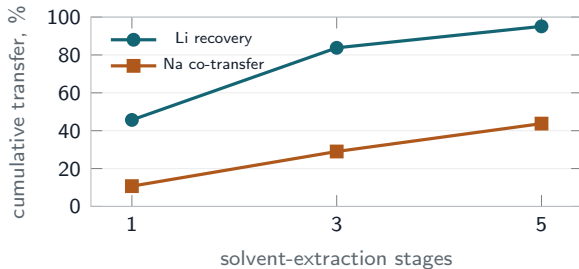


Rezaee extraction responses are converted to Li/Na transfer variables for process calculations

Stages	Li recovery	Na co-transfer
1	45.681%	10.714%
3	83.770%	29.008%
5	95.100%	43.740%

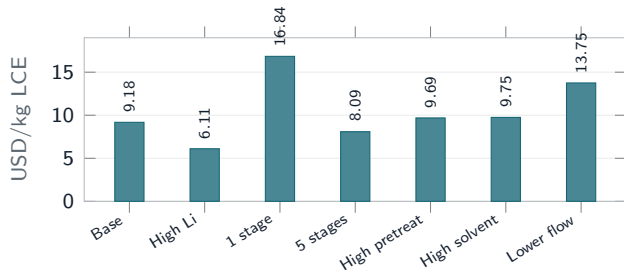
- The process layer uses PrOMMiS SolventExtraction objects with IDAES MSContactor units.
- Lithium and sodium material-transfer terms are active.
- Chloride transfer is held at zero inside the Li/Na extraction boundary.

Three Stages Recover 83.77% Lithium



- Smackover MS-2 at $O/A = 1$ and 10 wt% TOPO reaches 83.77% cumulative lithium recovery after three stages.
- Sodium co-transfer is 29.01% at the same condition.
- Five stages raise lithium recovery to 95.10% with a larger sodium burden.

Screening TEA Centers At 9.18 USD/kg LCE



- Base Smackover intensity is 9.18 USD/kg lithium carbonate equivalent.
- High-Li feed sensitivity lowers the value to 6.11 USD/kg.
- One stage raises the value to 16.84 USD/kg through lower recovery.

Economic Levers Are Feed Grade, Stages, And Losses

Scenario	Recovery	Cost	p10–p90
base Smackover	79.58%	9.18	8.81–12.43
high-Li Smackover	79.72%	6.11	5.92–8.26
one stage	43.40%	16.84	16.15–22.52
five stages	90.35%	8.09	7.77–10.74
high pretreatment loss	75.39%	9.69	8.93–12.57
high solvent loss	79.58%	9.75	9.00–12.59
lower throughput	79.58%	13.75	12.65–18.96

Cost values in USD/kg lithium carbonate equivalent.

Screening boundary

The TEA compares operating sensitivities for prioritization; it does not replace equipment design, residence-time sizing, solvent-loss experiments, or product-pricing analysis.

- Feed lithium grade is the strongest favorable lever.
- Stage count trades lithium recovery against sodium loading.
- Fixed annual charges dominate the lower-throughput case.

THERMODYNAMIC EXTENSION

Case-Study Coverage Matrix

Lane	Current basis	Process-facing use
feed basis	Smackover MS-2 and high-Li Smackover sensitivity	defines the main and feed-grade sensitivity cases
paper-response bridge	Rezaee Li/Na extraction responses with <i>O/A</i> conversion	supplies D_{Li} , D_{Na} , and $S_{Li/Na}$ for staged extraction
ALAMO surrogate	held-out parity for logit k_{Li} and logit k_{Na}	provides the compact transfer surface used by the process model
PrOMMiS/IDAES stages	PrOMMiS/IDAES staged-extraction terms for Li and Na across 1, 3, and 5 stages	converts transfer variables to outlet recovery and raffinate behavior
screening TEA/UQ	feed grade, stage count, pretreatment loss, solvent loss, and throughput	ranks operating levers for sprint approval
direct ePC-SAFT replacement path	accepted Li/Na transfer rows	replaces the paper-response bridge without changing process inputs

Process-facing rows remain stable as the thermodynamic basis is exchanged.

The case study is a screening-quality bridge from Smackover feed chemistry to PrOMMiS/IDAES staged extraction and TEA, ready to anchor an internal ePC-SAFT-to-PrOMMiS/IDAES integration sprint.

Technical Message

1. Smackover MS-2 is the main produced-water feed; the high-Li Smackover row defines the feed-grade sensitivity.
2. TBAC/DA DES + 10 wt% TOPO maps lithium and sodium transfer through ALAMO and into staged solvent extraction.
3. Three stages recover 83.77% lithium before the TEA pretreatment-loss adjustment.
4. The base screening TEA is 9.18 USD/kg LCE, with feed grade and stage count controlling the largest movement.
5. Ask: approve an internal ePC-SAFT-to-PrOMMiS/IDAES integration and validation sprint.

Supplemental Extension Context

Family	Why it belongs near produced water	Needed before modeling
CMM/REE ions	oilfield brines can carry critical-mineral value beyond Li	measured composition, aqueous speciation, and ion parameters
divalent ions	Ca, Mg, and Sr set pretreatment load and leakage risk	pretreatment data and extraction leakage bounds
hydrocarbons	oil-bearing water adds phase behavior and solvent-contact risks	oil-phase assay and phase-equilibrium data
ePC-SAFT extension	electrolyte and Born/dielectric terms give a thermodynamic route	accepted parameters and phase data for each species family

Boundary

Li/Na extraction remains the present case. CMM, REE, and hydrocarbon rows define where additional data would expand the same structure.

Use

The extension list guides what to measure before adding species, reactions, or oil-phase terms to the thermodynamic model.

USGS Energy Waste Science; critical-mineral produced-water review; electrolyte PC-SAFT permittivity and ePC-SAFT Born/dielectric literature.

Sources

- Rezaee et al. 2025: TBAC/DA DES + TOPO extraction response basis.
- Rezaee et al. 2026: PC-SAFT/ePC-SAFT parameter and reaction-equilibrium support.
- U.S. Geological Survey southern Arkansas Smackover brine data release, DOI: 10.5066/P9QPRYZN.
- USGS Energy Waste Science and produced-water lithium/mineral recovery context.
- Critical-mineral source potential review for U.S. oil and gas produced waters.
- Electrolyte PC-SAFT permittivity and ePC-SAFT Born/dielectric literature for the extension basis.
- Produced-water feed table, ALAMO validation table, PrOMMiS/IDAES staged-extraction table, and screening TEA table from the case-study result tables.